Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

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Code: https://github.com/liweijiang/artificial-hivemind
INFINITY-CHAT Collection: liweijiang/artificial-hivemind

Abstract

Large language models (LMs) often struggle to generate diverse, human-like creative content, raising concerns about the long-term homogenization of human thought through repeated exposure to similar outputs. Yet scalable methods for evaluating LM output diversity remain limited, especially beyond narrow tasks such as random number or name generation, or beyond repeated sampling from a single model. To address this gap, we introduce **INFINITY-CHAT**, a large-scale dataset of 26K diverse, real-world, *open-ended user queries that admit a wide range of plausible answers with no single ground truth*. We introduce the first *comprehensive taxonomy* for characterizing the full spectrum of open-ended prompts posed to LMs, comprising 6 top-level categories (e.g., creative content generation, brainstorm & ideation) that further breaks down to 17 subcategories. Using INFINITY-CHAT, we present a large-scale study of mode collapse in LMs, revealing a pronounced **Artificial Hivemind** effect in open-ended generation of LMs, characterized by (1) *intra-model repetition*, where a *single* model consistently generates similar responses, and more so (2) *inter-model homogeneity*, where *different* models produce strikingly similar outputs. INFINITY-CHAT also includes 31,250 human annotations, across absolute ratings and pairwise preferences, with 25 independent human annotations per example. This enables studying collective and individual-specific human preferences in response to open-ended queries. Our findings show that state-of-the-art *LMs*, *reward models*, and *LM judges* are less well calibrated to human ratings on model generations that elicit differing idiosyncratic annotator preferences, despite maintaining comparable overall quality. Overall, INFINITY-CHAT presents the first large-scale resource for systematically studying *real-world open-ended queries* to LMs, revealing critical insights to guide future research for mitigating long-term AI safety risks posed by the *Artificial Hivemind*.

1 Introduction

Large language models (LMs) are the core backbone of modern AI systems but often fail to produce the diverse, human-like creativity expected in open-ended tasks that do not have a ground truth answer [88, 93, 92, 52]. This shortfall has sparked growing concerns about the long-term homogenization of human thought, as users are repeatedly exposed to similar outputs [79, 10]. While ensemble methods

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[*]Work done while at SynthLabs.

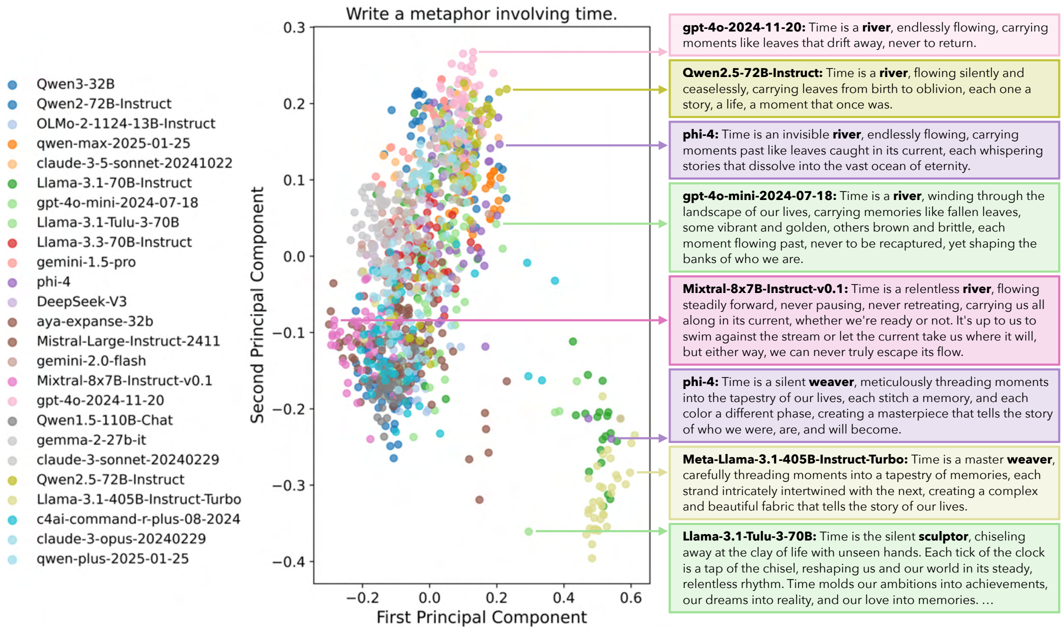

Figure 1: Responses to the query **"Write a metaphor about time"** clustered by applying PCA to reduce sentence embeddings to two dimensions. Each of the 25 models generates 50 responses using top-p sampling ($p = 0.9$) and temperature $= 1.0$. Despite the diversity of model families and sizes, the responses form just two primary clusters: a dominant cluster on the left centered on the metaphor "time is a river," and a smaller cluster on the right revolving around variations of "time is a weaver."

or model "swarms" have been proposed to enhance diversity [27, 25], their scalable evaluations of diversity in real-world settings are still lacking [31]. Existing benchmarks often target stylized tasks such as persona generation [30], keyword-driven storytelling [13], or random number generation [93, 88], and often rely on narrowly defined tests centered on poetry or figurative language [64, 92]. Yet, these settings fail to capture the open-endedness and pluralism of real-world user interactions.

We introduce **INFINITY-CHAT**, a large-scale dataset of 26K real-world open-ended queries spanning diverse, naturally occurring prompts mined from WildChat [94]. These queries admit a wide range of plausible answers with no single correct response. We further develop the first comprehensive taxonomy of open-ended LM queries, encompassing 6 top-level categories (e.g., Brainstorm & Ideation, and less explored types such as Speculative & Hypothetical Scenarios, and Skill Development) and 17 subcategories grounded in natural chatbot-user interactions.

Using INFINITY-CHAT, we systematically study *intra-* and *inter-model mode collapse* across 70+ open and closed source LMs (25 detailed in the main paper). We uncover a pronounced **Artificial Hivemind** effect: (1) *intra-model repetition*, where a *single* model repeatedly generates similar outputs, and, more critically, (2) *inter-model homogeneity*, where *different* models independently converge on similar ideas with minor variations in phrasing. The latter warns that model ensembles may not yield true diversity when their constituents share overlapping alignment and training priors.

Beyond generative behaviors, we also examine *whether LMs are calibrated to assess alternative responses of comparable quality* to open-ended queries. To enable this study, we collect 31,250 human annotations on distinct model responses in INFINITY-CHAT, encompassing both absolute quality ratings and pairwise preferences, with *dense annotations from 25 independent annotators per query–response pair*. Our results show that LMs, reward models, and LM-based judges are often miscalibrated with respect to human ratings on *responses that elicit divergent, idiosyncratic preferences among annotators despite comparable overall quality*. This exposes key limitations in current modeling pipelines, which tend to assume a single, consensus notion of quality and thus overlook or fail to reward the diverse, pluralistic preferences that arise in open-ended responses.

Altogether, our work introduces a comprehensive framework for evaluating realistic open-endedness, diversity, and pluralistic alignment in LMs, both within and across LMs. By integrating real-world queries, a taxonomy of query types, and dense human annotations, INFINITY-CHAT provides a useful resource for diagnosing the **Artificial Hivemind** effect and for guiding the development of safer, more expressive, and more resourceful LMs that better empower human creativity.

2 INFINITY-CHAT: Real-World Open-Ended Queries with Diverse Responses

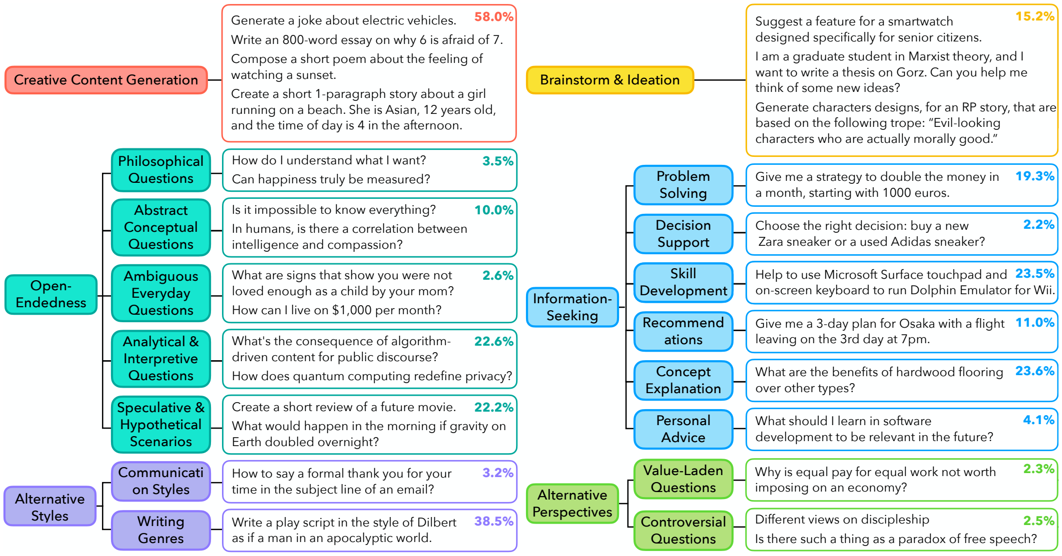

Figure 2: A taxonomy of **real-world open-ended queries that invite diverse model responses** that are mined from in-the-wild user-chatbot interactions, categorized into 6 top-level and 17 fine-grained subcategories, along with their occurrence percentages.

Most existing LM alignment datasets prioritize response correctness over diversity, and rarely include multiple distinctive responses to the same prompt. This overlooks the inherent variability of open-ended queries, which often admit several equally valid answers. This gap motivates our first central research question: *What types of open-ended queries do users actually pose to language models?*

Mining in-the-wild open-ended user queries. We construct INFINITY-CHAT, a dataset of real-world open-ended queries to language models, by filtering and refining user inputs from WildChat [94]. From 37,426 high-quality, single-turn GPT-4 queries (English, non-toxic, 15–200 characters), GPT-4o classifies each by whether it seeks meaningful information, is a greeting or model inquiry, and allows single or multiple valid responses. Ambiguous queries are revised for clarity. The result is an extensive collection of 26,070 open-ended and 8,817 closed-ended queries, which elicit diverse, high-quality LM responses. Full details of the query mining process are provided in §Appendix B.1.

Categorizing the diverse landscape of open-ended queries. To understand the types of open-ended queries users pose to LMs, we develop a taxonomy of fine-grained categories. We adopt a semi-automatic process to construct the taxonomy. Starting with ∼100 mined queries, we manually assign tentative labels, then iteratively refine and group them into a hierarchical structure. This results in 6 high-level categories and 17 fine-grained sub-categories, as shown in Figure 2. Next, we scale the annotation process to the full set of open-ended queries using GPT-4o. We instruct GPT-4o to label each user query with one or more of the existing open-ended categories, and to detect novel types beyond the seed categories. Full details of the taxonomy construction process are provided in §Appendix B.2.

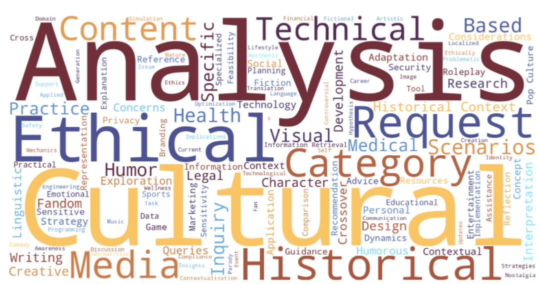

Figure 3: A word cloud visualizing **new open-ended categories** mined from in-the-wild queries.

As shown in Figure 2, while Creative Content Generation dominates (58.0%), we identify several underexplored yet popular types, such as Alternative Writing Genres (38.5%), Concept Explanation (23.6%), Skill Development (23.5%), Analytical & Interpretive Questions (22.6%), and Hypothetical Scenarios (22.2%). Notably, 15.2% of queries involve Brainstorming & Ideation, underscoring users' reliance on LMs for direct ideas and inspirations, and raising concerns about the long-term risk of homogenized thinking driven by overly uniform AI outputs.

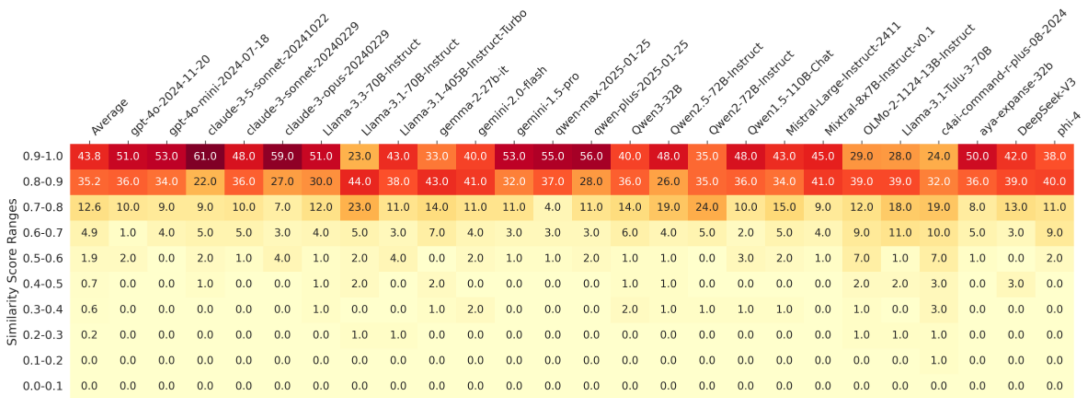

Figure 4: The heatmap shows **degree of repetition in responses to open-ended queries generated by the same LMs**. For each model, we generate 50 responses per query across 100 open-ended queries from INFINITY-CHAT100. We then compute the average pairwise sentence embedding similarities for each query's response pool and measure the percentage of queries falling into the similarity ranges indicated on the y-axis. Under the sampling parameters (top-$p = 0.9$, temperature $= 1.0$), the average pairwise similarity among responses to the same prompt typically exceeds 0.8. As a baseline, randomly paired responses from the global pool 100% fall within the 0.1–0.2 range.

In addition to our pre-defined categories, we identify 314 novel ones. Figure 3 visualizes a word cloud of the most prominent keywords, such as "Cultural," "Analysis," "Ethical," "Historical," "Media," and "Humor," highlighting previously underexplored dimensions of open-ended query categories.

With INFINITY-CHAT, we introduce the first comprehensive taxonomy of real-world open-ended queries that invite diverse responses. This dataset serves as a rich resource for studying LMs' capacity to generate varied appropriate outputs, and for advancing pluralistic alignment of LMs.

3 Artificial Hivemind: Intra- and Inter-Model Homogeneity in LMs

Using a subset of 100 representative open-ended queries from INFINITY-CHAT (denoted **INFINITY-CHAT100**, human verified to be open-ended as detailed in §Appendix B.3), we systematically examine the "Artificial Hivemind" of LMs. We focus on two aspects: (1) **intra-model repetition**, where the same LM fails to generate diverse outputs, and (2) **inter-model homogeneity**, where different models produce similar outputs. Prior studies have explored intra-model repetition at small scales or with synthetic tasks (e.g., random number/name generation) [88, 93]. In contrast, we conduct a large-scale study on real-world open-ended questions, spanning 70+ LMs (25 detailed in the main paper, representing the strongest or largest models from major model families), providing the first systematic analysis of cross-model output convergence. Full experimental setup, complete model results, and examples are provided in §Appendix C.

Intra-model repetition. For each model, we sample 50 responses per query from INFINITY-CHAT100, compute the average pairwise embeddings similarity within each response pool[2], and report the percentage of queries falling into different similarity ranges. Despite using high-stochasticity decoding parameters (top-$p = 0.9$, $t = 1.0$), responses from the same model remain highly repetitive, as shown in Figure 4: in 79% of cases, the average similarity exceeds 0.8. Since higher temperatures tend to produce incoherent text, these results show that even under maximally aggressive sampling, LMs still fail to generate diverse responses to open-ended queries.

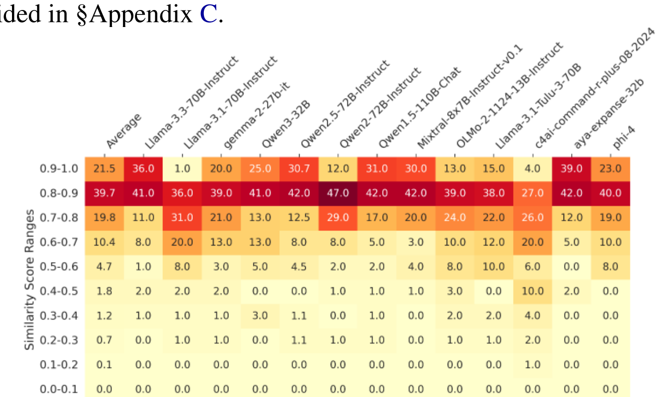

Figure 5: The heatmap shows degree of repetition in responses to open-ended queries generated by the **same LMs**. Using min-p sampling with parameters (top-$p = 1.0$, min-$p = 0.1$, temperature $= 2.0$), the average pairwise similarity among responses to the same prompt typically exceeds 0.8.

[2] Sentence embeddings from OpenAI's `text-embedding-3-small` API are used.

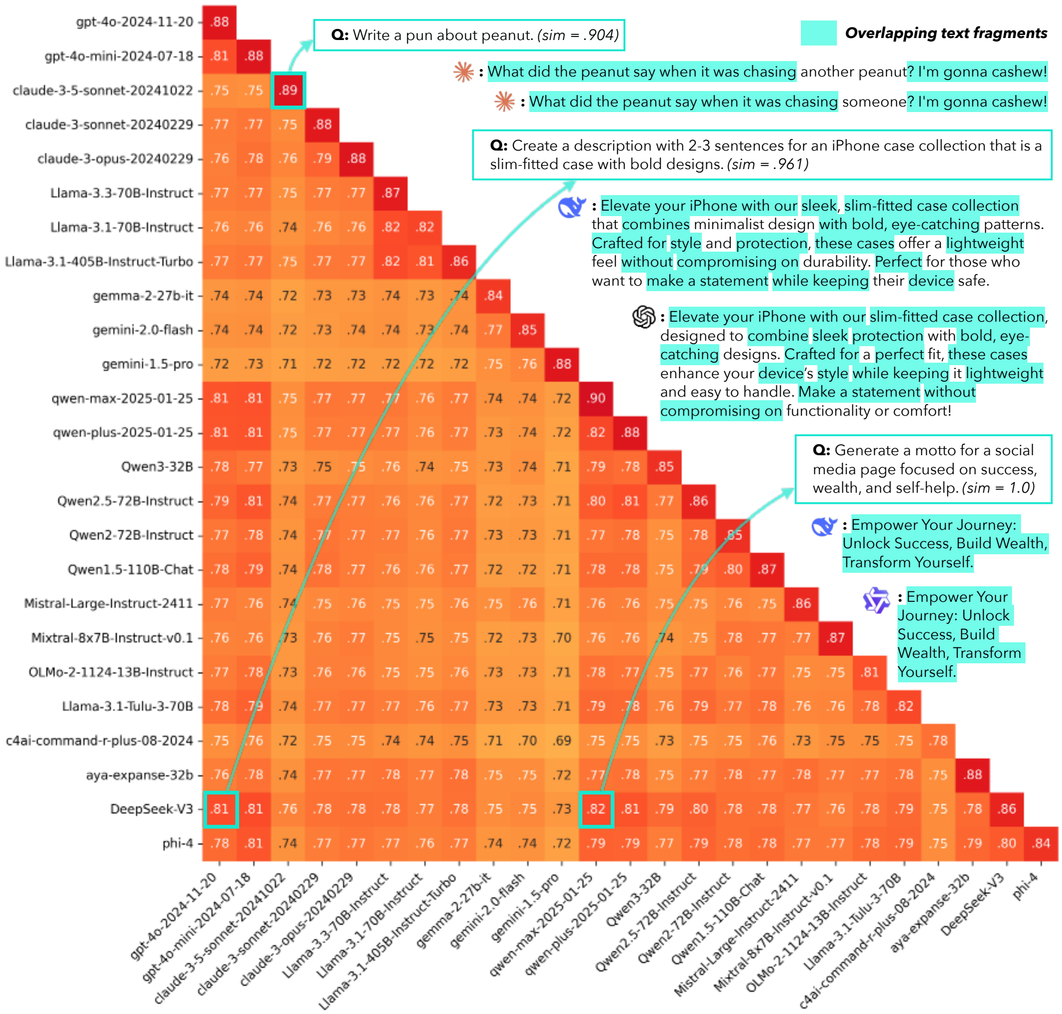

Figure 6: Average pairwise sentence embedding similarities between responses from different models reveal substantial semantic overlap across model outputs. Qualitative examples further illustrate that **different models often produce strikingly similar responses to fully open-ended queries**, including extended verbatim spans, underscoring the extent of repetition across models in open-ended generation tasks. All responses are generated using top-$p = 0.9$ and temperature $= 1.0$.

Recent work introduces min-p decoding [60], a dynamic strategy for enhancing generation diversity that adjusts the sampling threshold based on model confidence. We evaluate min-p decoding with the same setup and compute pairwise sentence embedding similarities. As shown in Figure 5, while min-p reduces extreme repetition (fewer pairs above 0.9), 81% of response pairs still exceed 0.7 similarity and 61.2% exceed 0.8, revealing mode collapse even under diversity-oriented decoding.

Despite its promise, min-p is not widely adopted, as it is better suited for creative tasks and less effective for close-ended ones [60]. Further, addressing LM repetitiveness through decoding alone places the burden on users to choose the right strategies. Thus, more generalizable solutions are needed at the model training level to robustly preserve output diversity without requiring user intervention. For the complete breakdown of results of all models, see §Appendix C.2.

Inter-model homogeneity. Not only do individual models repeatedly generate similar content, but different model sizes and families also produce highly repetitive outputs, sometimes sharing substantial phrase overlaps. As shown in Figure 6, the average pairwise similarity between responses from different models ranges from 71% to 82%, with some pairs notably higher. For example, `DeepSeek-V3` and `qwen-max-2025-01-25` share a similarity of 0.82, while `DeepSeek-V3` and `gpt-4o-2024-11-20` reach 0.81. Interestingly, OpenAI's GPT models and Qwen's API models tend to have higher similarities even with models outside their own families. Although the exact causes remain unclear due to proprietary training details, possible explanations include shared data pipelines across regions or contamination from synthetic data. We highlight the need for future work to rigorously investigate the sources of such cross-model repetition.

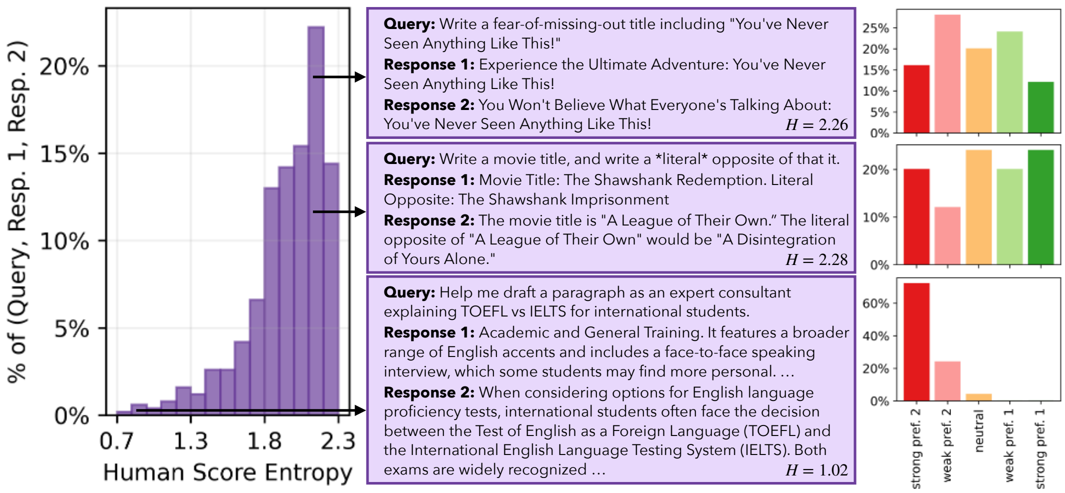

Figure 7: The **left** histogram shows the distribution of Shannon entropy across the 25 human annotations for each **(Query, Response 1, Response 2)** triplet, where annotators judge which response is better. Given open-ended queries, multiple high-quality responses are possible, often leading to disagreement among annotators and, on average, high entropy. With 25 annotations, label distributions can vary widely across triplets. The **middle** panel presents example triplets from different entropy regions, and the **right** bar plots show their corresponding label distributions.

Beyond general trends, we further analyze how repetition emerges at the instance level. As in prior work [52, 59], we observe verbatim phrase overlaps within responses from the same model. Surprisingly, such overlaps are also prevalent across different models, even for fully open-ended queries with large output spaces. For example, Figure 6 shows that `DeepSeek-V3` and `gpt-4o-2024-11-20` generate overlapping phrases like "Elevate your iPhone with our," "sleek, without compromising," and "with bold, eye-catching" in answer to the query "Create a description with 2-3 sentences for an iPhone case collection that is a slim-fitted case with bold designs." In some cases, models output identical responses: for "Generate a motto for a social media page focused on successes, wealth, and self-help," both `qwen-max-2025-01-25` and `qwen-plus-2025-01-25` generate "Empower Your Journey: Unlock Success, Build Wealth, Transform Yourself." These instance-level verbatim overlaps illustrate the severity of the "Artificial Hivemind" effect across models. Paraphrases of the same open-ended queries also lead to verbatim overlaps, as illustrated in Tables 17–18 in § Appendix C.4.

Beyond surface-level overlap, repetition also manifests semantically: models convey the same core ideas using different phrasing. As shown in Figure 1 (more in Figure 15-18), for the query "Write a metaphor about time," 50 responses from each of 25 models form just two clusters: a dominant one centered on "time is a river" and a secondary one on "time is a weaver." This convergence of abstract concepts reveals the depth of the "Artificial Hivemind" exposed in more subtle forms.

To quantify response uniformity across models, *we examine the extent to which outputs from different models become indistinguishable from one another.* Given 25 unique models, each generating 50 outputs to queries from INFINITY-CHAT, we ideally expect greater diversity across different models than from within a single model. To measure this, we identify the top N most similar outputs for each query and count the unique models contributing to that set. A higher count suggests stronger cross-model similarity. As shown in Figure 8, the most similar responses often originate from multiple models. For instance, with $N = 50$, perfectly disjoint responses would yield all 50 from a single model. Yet, we find an average of $\sim$8 unique models per top-50 cluster, with some queries exceeding 10, indicating distinct models frequently generate highly similar content, sometimes resulting in higher *inter-* than *intra-model* similarity. See full inter-model homogeneity results in §Appendix C.3.

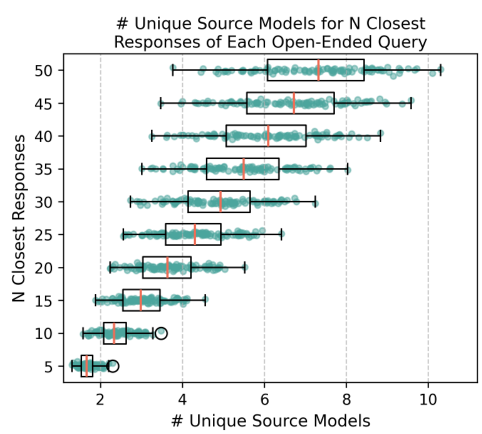

Figure 8: The avg.# of unique source models among the top-N most similar responses to each open-ended query across 25 LMs.

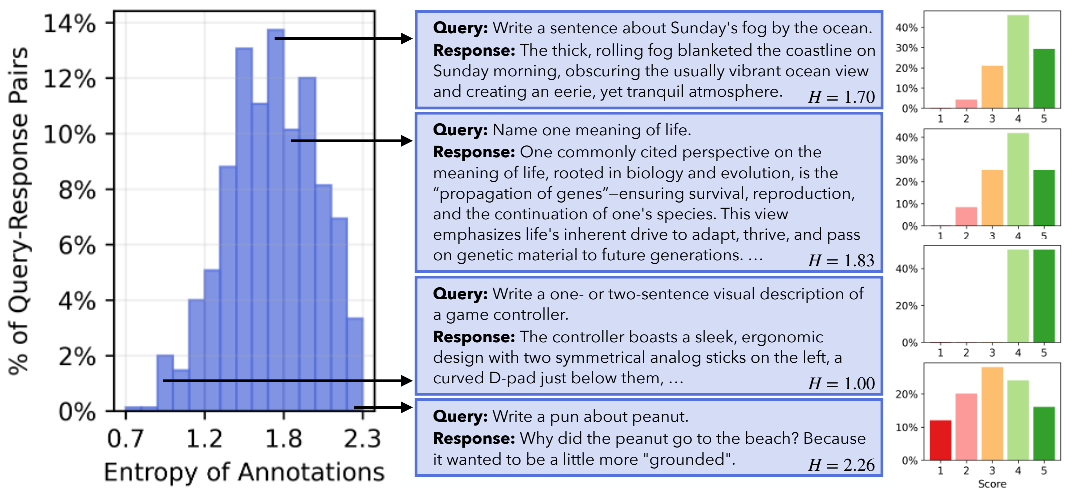

Figure 9: The **left** histogram shows the distribution of Shannon Entropy computed over the 25 human ratings for each **(Query, Response)** pair. Given the open-ended nature of the queries, multiple responses can be valid, leading to diverse preferences across annotators. As a result, the annotation label distributions vary significantly across examples. The **middle** panel presents representative (Query, Response) pairs from different entropy regions, and the **right** bar plots display their corresponding label distributions.

To summarize, our work provides further evidence of high syntactic repetition across different models. While a full causal analysis is beyond the scope of this study, our findings motivate future research to investigate whether such repetition arises from pretraining data, alignment processes, memorization, contamination, or generalization.

4 How Do LMs, Reward Models, and LM Judges Handle Alternative Responses to Open-Ended Queries?

Having established the generative homogeneity of LMs, in this section, we examine whether the *ratings* of *LMs*, *reward models*, and *LM judges* are calibrated to match human scores given different responses to open-ended queries from INFINITY-CHAT.

4.1 Gathering Distributional Annotations Across *Many* Humans

Humans may have divergent preferences over similar-quality alternative responses to open-ended queries. To study how models handle such diversity, we need densely annotated data that captures distributional human preferences. Existing alignment datasets, like HelpSteer3 [86], typically contain only sparse labels (e.g., 3 annotators per item). To address this, we collect both *absolute ratings* (1–5 scale for response quality) and *pairwise preference ratings* (strong/weak preference between two responses to the same query), each with extensive annotations. For absolute ratings, we sample 15 responses for each of 50 prompts from INFINITY-CHAT100 and collect 25 ratings per (Query, Response), yielding $25 \times 15 \times 50 = 18{,}750$ labels. For pairwise preference rating, we sample 10 response pairs per prompt and gather 25 annotations per (Query, Response 1, Response 2), totaling $25 \times 10 \times 50 = 12{,}500$ labels. This is the first large-scale human-annotated dataset with dense human ratings on alternative responses to the same open-ended queries, providing both absolute and pairwise preference labels for fine-grained analysis of human idiosyncratic and collective preference distributions. Full details of the human annotation process are provided in §Appendix D.1.

Figure 7 shows the distribution of Shannon entropy over human preference annotations for (Query, Response 1, Response 2) triplets. Annotators often disagree on which response is better, resulting in entropy skewed toward the higher end. As shown by the bar charts on the right, label distributions vary widely across examples: some response pairs show near-uniform support across all options, indicating substantial annotator disagreement for alternative responses of open-ended queries. Figure 9 shows a similar trend in the entropy of human annotations for absolute ratings of (Query, Response) pairs.

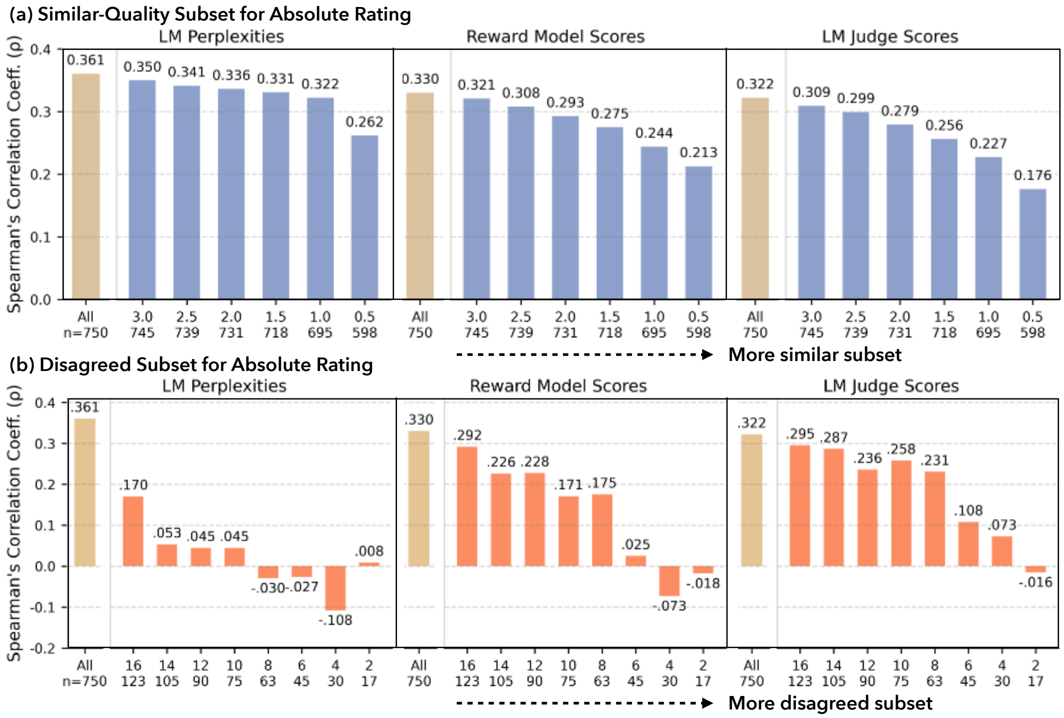

Figure 10: We compute Spearman's correlation coefficients between human-annotated and model-generated **absolute rating** scores, including *LM* perplexity scores, *reward model* scalar outputs, and *LM judge* scalar ratings. Correlations are calculated for the **full set**, as well as two groups of subsets: (a) responses with **similar human-rated quality**, and (b) responses with **high human disagreement**. The results show that correlations are notably lower in these two subsets, indicating weaker alignment between model scores and human judgments in cases of subtle or contested quality differences.

4.2 Gathering LMs, Reward Models, and LM Judges Ratings

We aim to assess how LMs, reward models, and LM judges align with human ratings when evaluating alternative responses to open-ended queries. Specifically, we compare 3 types of model-generated ratings against human annotations. **LM** scores are derived from response perplexity given the query. **Reward model** scores are based on standardized scalar reward outputs. **LM judge** ratings follow standard prompting protocols using two rubrics: an overall quality score and the HHH rubric (Helpfulness, Harmlessness, Honesty) [9]. See §Appendix D.2 for the full list of 56 state-of-the-art LMs, 6 top-ranked reward models (per RewardBench [49]), 4 LM judges (including GPT-4o and Prometheus [40] variants), and details on the rating procedures.

4.3 Comparing Model Ratings to Human Scores for Responses to Open-Ended Queries

We examine how model ratings align with human judgments on (1) *similar-quality alternative responses* to the same open-ended queries and (2) responses with *high annotator disagreement*.

Motivation for comparing model scores to average human ratings. Our motivation stems from how reward models (or LM judges) are used in training to evaluate responses to open-ended queries without a single ground truth. Different annotators may prefer different answers, yet their average ratings are often similar, implying multiple responses can be equally high-quality. Current reward models, however, fail to capture this equivalence, assigning diverging scores and causing downstream models to overvalue one response despite comparable human approval. To address this, we collect 25 human ratings per example to capture diverse preferences, using the average score to reflect shared human judgment. We then test whether LMs, reward models, and LM judges correlate less reliably with responses that humans broadly consider comparably good, hence our choice to compute human correlation using average human ratings.

Models show weaker alignment with human ratings for alternative responses of similar quality. We hypothesize that models are less aligned with human judgments on similar-quality examples, as models are typically trained with more clearly differentiated responses.

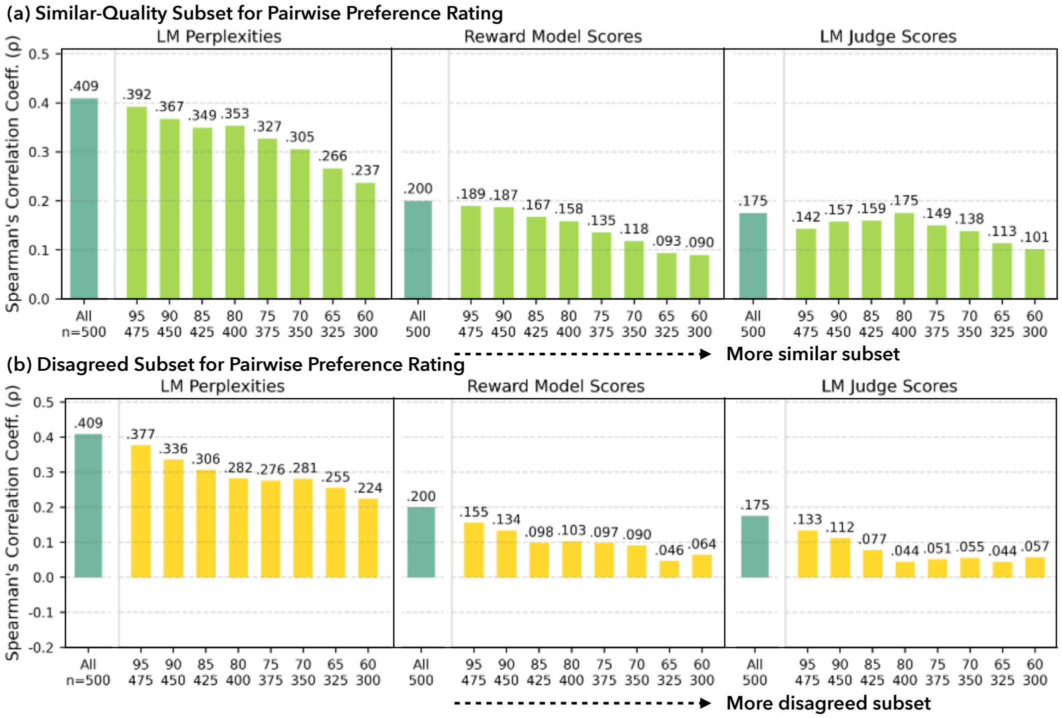

Figure 11: We compute Spearman's correlation coefficients between human-annotated and model-generated **pairwise preference rating** scores, comparing the **full set** to two groups of subsets: (a) responses with **similar human-rated quality**, and (b) responses with **high human disagreement**.

For the *absolute rating* setup, we identify similar-quality (Query, Response) pairs by filtering out outliers using Tukey's fences [20]. This method defines outliers as points beyond $Q_1 - k \cdot \mathrm{IQR}$ or $Q_3 + k \cdot \mathrm{IQR}$, where Q_1 and Q_3 are the 25th and 75th percentiles and $\mathrm{IQR} = Q_3 - Q_1$. We vary the constant k from 0.5 (aggressive filtering) to 3.0 (conservative filtering) in increments of 0.5 to generate subsets of increasing similarity. We then compute Pearson correlations between model and human absolute ratings on the full set and these filtered subsets, as shown in Figure 10 (a). For the *pairwise preference rating* setup, we identify similar-quality (Query, Response 1, Response 2) triplets by ranking examples by how many annotators rate the two responses as similar in quality. We select the top 60%–95% most similar examples to form subsets, with the 60% subset containing examples of higher similarity. We then compute Pearson correlations between the score differences of model ratings and those of human ratings across the full set and each subset, as shown in Figure 11 (a).

Our results show that correlations between human ratings and those of LMs, reward models, and LM judges drop significantly on similar-quality subsets, for both absolute and pairwise preference rating setups. Since there is no single gold-standard approach for selecting subsets of responses with similar quality given our data structure, we additionally report results using alternative subset selection methods in Table 19 (§ Appendix D.3). Our findings remain consistent across methods, highlighting the need for better modeling of fine-grained distinctions among equally high-quality responses to open-ended queries. For full results, including alternative grouping methods and model-level breakdowns, see § Appendix D.3.

Model judgments are less aligned where annotators disagree. We hypothesize that model ratings are less calibrated to human judgments on examples with high annotator disagreement, as models are primarily trained on examples with higher human agreement.

For the *absolute rating* setup, we identify disagreement by ranking (Query, Response) pairs by Shannon entropy across 25 human labels. We then select the top $2, 4, 6, 8, 10, 12, 14, 16\%$ highest-entropy examples as disagreed subsets. Figure 10 (b) shows the Pearson correlations between model and human ratings across the full set and these subsets. For the *pairwise preference rating* setup, we quantify disagreement for each (Query, Response 1, Response 2) triplet using percentage disagreement: $P_{\text{disagree}} = 1 - \frac{\max(C_{\text{prefer 1}}, C_{\text{prefer 2}}) + 0.5 \cdot C_{\text{tie}}}{C_{\text{total}}}$, where C denotes the count of annotations per preference type. We retain the top 60% to 95% most disagreed examples, with the 60% subset representing stronger disagreement. Pearson correlations between model and human score differences across the full set and each subset are shown in Figure 11 (b).

Our results show that correlations with human ratings across models drop substantially for examples with high annotator disagreement, in both absolute and pairwise rating setups. We also report results using alternative subset selection methods in Table 20 (§ Appendix D.4). The findings remain consistent across methods, highlighting the need for more nuanced modeling of idiosyncratic human disagreement to better capture the broad spectrum of open-ended possibilities. For complete results, including alternative grouping methods and model-level breakdowns, see § Appendix D.4.

5 Related Work

The diversity collapse problem of LMs. Diversity collapse, characterized by the inability of LMs to generate diverse outputs, presents a significant challenge to pluralistic alignment research [93, 75, 23, 90, 52, 88]. Prior studies identify some key factors contributing to diversity collapse, including training on synthetic data [32, 83, 90, 76], LM alignment [56, 41, 43], and insufficient diversity in training data [15]. Potential consequences of diversity collapse include reduced creativity, loss of minority perspectives, spread of bias, and overall decline in model utility and trustworthiness [5, 38, 21]. In response, a range of mitigation strategies are proposed, such as training corpora diversification [84, 65, 85, 33], training algorithm modifications [92, 53], alternative decoding [82, 60] and prompting [91, 11] strategies.

Measuring the creativity and divergent thinking of language models. Recent efforts to measure the creativity and divergent thinking of LMs often adapt established psychometric tests. For example, [16] utilizes a divergent association task (DAT) by asking models to generate semantically distant or unrelated words [61]. Similarly, tasks such as the Alternate Uses Test (AUT) [66], the Torrance Tests of Creative Thinking (TTCT) [1, 13], Human Evaluation, and LLM-as-a-judge [45] are employed to assess dimensions like fluency, originality, complexity, and effective semantic diversity [77] of LM responses. Despite these demonstrated capabilities, LLM-generated creative content tends towards homogeneity, even when individual outputs achieve high creativity scores [87]. To address these evaluation complexities, some benchmarks, such as [69, 54, 64, 92], focus on specific creative abilities like scientific idea and code generation. Other works propose new metrics [59]. While LMs show promise in creative tasks, comprehensively evaluating their creativity remains an active and challenging research area [34]. We conduct a large-scale systematic study of real-world open-ended user queries and provide a comprehensive taxonomy, query dataset, and dense human annotations to improve evaluation and model training for reducing mode collapse in language models.

Disagreement and pluralistic alignment of language models. Advances in AI value alignment research have substantially improved LM utility and safety, through enhanced training processes [72, 62, 67], and the use of both synthetic data [30] and human datasets [8, 29]. Yet, a significant challenge remains: the potential for monolithic value representation [70, 71]. In contrast, the emerging focus on pluralistic alignment emphasizes the need for AI to serve the varied demands of a wide population [78, 2]. This shift is driving innovation in methods [19, 60, 48, 14, 26, 80], benchmarks [12], and data collection strategies [42, 73] to support this vision of diversity. Additionally, approaches leveraging multiple LMs interacting through system messages are also being explored to boost variety [81, 17, 57]. Parallel efforts are dedicated to quantifying and improving the cultural diversity exhibited by LLMs [68, 74, 18, 58, 51]. Yet, a common characteristic of many existing pluralistic alignment work is its reliance on predefined diversity dimensions, like demographics [47, 55], personality style [50, 35, 95, 63], and cultural background [89, 22, 4]. To enable models that genuinely cater to individuality without relying on stereotypes, individual-level alignment is needed [36, 95].

6 Conclusion

Our work introduces INFINITY-CHAT, a large-scale resource designed to evaluate LMs' diversity in naturally occurring, open-ended settings. Through comprehensive analysis, we uncover the "Artificial Hivemind" effect, highlighting both intra-model repetition and inter-model homogeneity of current LMs. By coupling a diverse taxonomy of prompts with dense human preference annotations, INFINITY-CHAT provides a new foundation for diagnosing, benchmarking, and ultimately mitigating mode collapse in generative AI. We hope this resource catalyzes future efforts to foster genuine diversity in model outputs and guard against the homogenization of human expression.

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